

# In-class template

An Bui

## 1. Set up

### Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

### Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
```

```

aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects"
)

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+")
)

```

## 2. Cleaning

### sites object

Goal: create a new object from `sites` that only includes NCOS sites

```

# creating new object from sites
ncos_sites <- sites |>
  # filtering to only include NCOS quarterly sampling sites
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

### clean taxon\_list

Goal: convert character strings in the existing `taxon_list` data frame into snakecase

```

# creating new object from taxon_list
taxon_list_clean <- taxon_list |>
  # converted taxon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))

```

## clean water\_quality

Goal: create a data frame with median pH, median DO (mg/L), median salinity (ppt) for each date and site

- create a new object from `water_quality`
- clean column names
- only include surveys from quarterly NCOS sites
- create a new column called `date_site` to join with later data frames
- group by `date_site`
- calculate median pH, salinity, and dissolved oxygen
- ungroup data frame
- filter out any outliers

```
# creating a new object from water quality
water_quality_clean <- water_quality |>
  # cleaning column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # creating new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # grouping by date/site
  group_by(date_site) |>
  # calculating median pH, salinity, and DO
  summarize(
    med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungrouping the data frame
  ungroup() |>
  # filtering out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

## clean aq\_ins

Goal: create a long format data frame in which each row contains the counted individuals in each taxon during each survey. The data frame should include taxonomic information for each organism and the water quality information for each survey

- create a new object from `aq_ins`
- clean column names
- filter to only include quarterly NCOS sites

- select columns
- filter to only include “filtered beaker” sample
- convert the data frame to long format
- group by site, date, and taxon
- calculate average abundance per liter (sum counts divided by 7.5 liters)
- create a new column called `date_site` to join with water quality
- join data frames
- join with taxon list

```
# create a new object
aq_ins_clean <- aq_ins |>
# cleaning column names
clean_names() |>
# filtering join: only include quarterly NCOS surveys from ncos_sites
semi_join(ncos_sites, by = "site") |>
# renaming column
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filtered to only include filtered beaker samples
filter(sample_type %in% c("FB250", "FB 250")) |>
# make data frame longer
pivot_longer(
  # columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for the old column names
  names_to = "taxon",
  # new column for the values in each of the old columns
  values_to = "abundance"
) |>
# grouped by date, site, and taxon
group_by(date, site, taxon) |>
# calculated average count/liter (total abundance/7.5 L)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
```

```

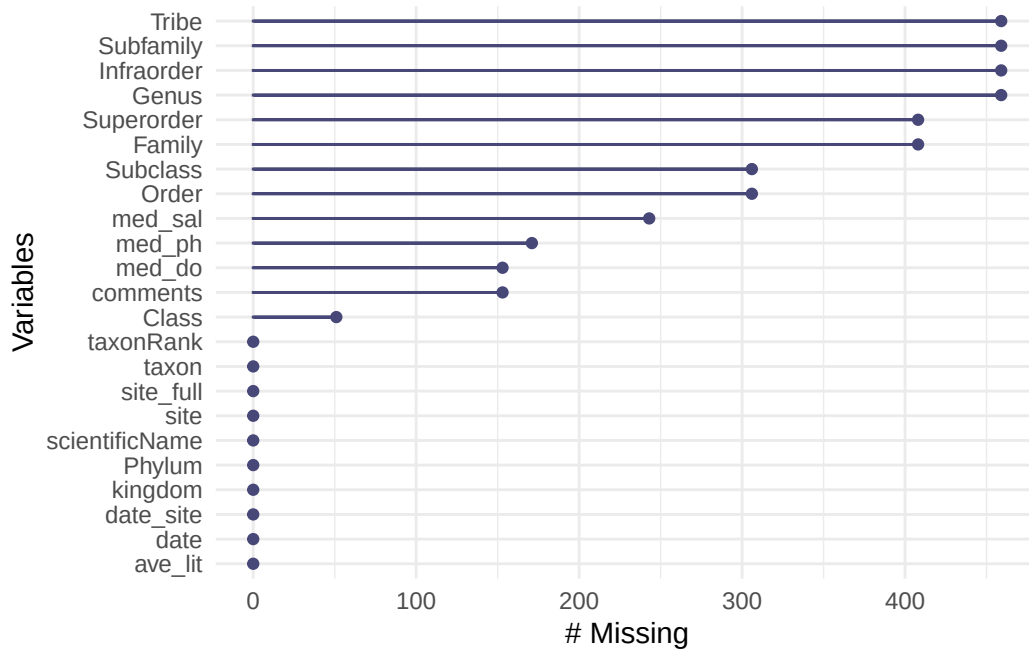
# ungrouped
ungroup() |>
# created new column to join with data frames
unite("date_site", date, site, remove = FALSE) |>
# joined with water quality clean
left_join(water_quality_clean, by = "date_site") |>
# joined with taxonomic information
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# fill in full site names
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "East Channel",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for full site name by median salinity
mutate(site_full = fct_reorder(site_full,
                               med_sal,
                               .fun = "median",
                               .na_rm = TRUE),
       site_full = fct_rev(site_full))

```

### 3. Visualizing

#### Visualize missing values

```
gg_miss_var(aq_ins_clean)
```



Missing values for salinity, pH, and dissolved oxygen might influence later figures.

## Visualize salinity

- x-axis: `site_full`
- y-axis: `mean_sal`
- geometry to show salinity during survey: `geom_jitter`
- geometry to show median salinity: `stat_summary`

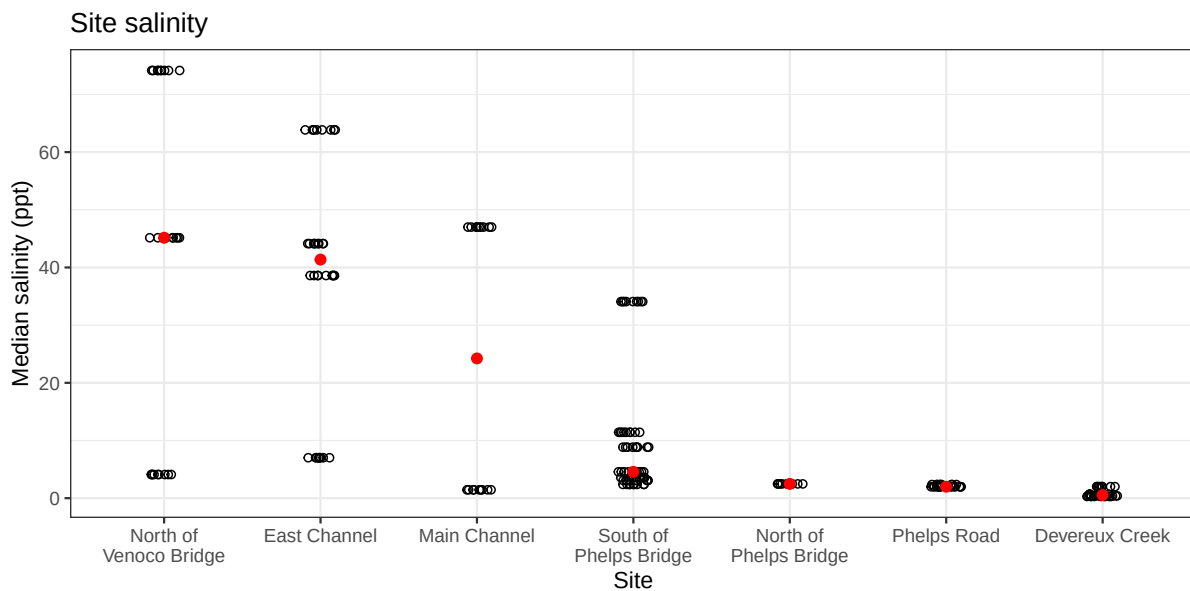
```
# creating a new object to store plot
salinity_plot <- ggplot(data = aq_ins_clean,
                        # x-axis: full site name
                        # y-axis: median salinity
                        mapping = aes(x = site_full,
                                      y = med_sal)) +
  # points representing median salinity at each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # points representing median salinity
  stat_summary(geom = "point",
              fun = median,
              color = "red",
```

```

        size = 2) +
# wrapping x-axis labels
scale_x_discrete(labels = label_wrap(width = 15)) +
# relabelling axes, adding title
labs(x = "Site",
     y = "Median salinity (ppt)",
     title = "Site salinity") +
# using a different theme from default
theme_bw()

# displaying the object
salinity_plot

```



Sites seem to differ in salinity. North of Venoco Bridge, East Channel, and Main Channel tend to be hypersaline, while North of Phelps Bridge, Phelps Road, and Devereux Creek is the least saline. South of Phelps Bridge tends to be more freshwater but is higher in salinity than the other, lower salinity sites.

### Visualize differences in copepod abundance

- x-axis: `site_full`
- y-axis: `ave_lit`
- geometry to represent observations: `geom_jitter()`
- geometry to represent medians: `stat_summary()`

## Initial cleaning:

```
copepods <- aq_ins_clean |>
  filter(taxon == "copepod") |>
  mutate(log_ave_lit = log(ave_lit + 1))
```

## Visualizing:

```
# creating object to store plot
copepod_plot <- ggplot(data = copepods,
  # x-axis: full site name
  # y-axis: log transformed average count/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +

# jitter to represent each sample
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +

# points to represent median copepod abundance
stat_summary(geom = "point",
  fun = median,
  color = "blue",
  size = 2) +

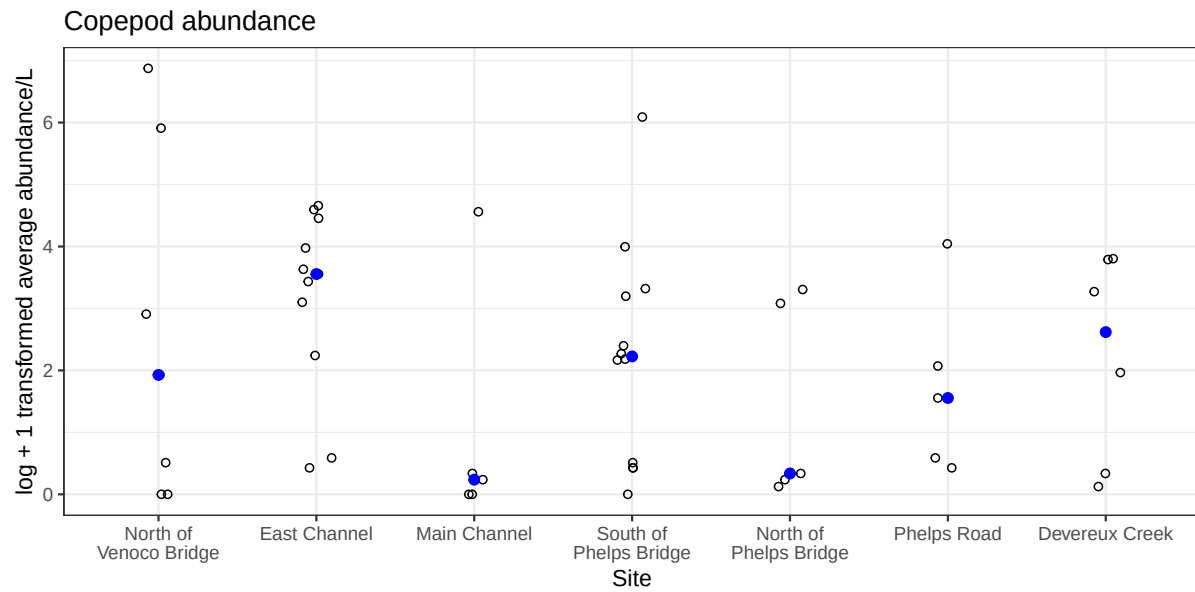
# wrapping x-axis labels
scale_x_discrete(labels = label_wrap(width = 15)) +

# relabelling axes
labs(x = "Site",
  y = "log + 1 transformed average abundance/L",
  title = "Copepod abundance") +

# using different theme
theme_bw()

# displaying the object
copepod_plot
```





Copepod abundance differs across sites.

### Putting salinity and copepod abundance together

```
salinity_plot / copepod_plot
```

The map displays the study area with various sampling sites marked by colored dots. The sites are categorized by location: North of Venoco Bridge, East Channel, Main Channel, South of Phelps Bridge, North of Phelps Bridge, Phelps Road, and Devereux Creek. The map includes a grid and a legend for the sampling sites.

Scatter plot showing the log + 1 transformed average abundance/L for various sites. The y-axis ranges from 0 to 6. The x-axis lists seven sites: North of Venoco Bridge, East Channel, Main Channel, South of Phelps Bridge, North of Phelps Bridge, Phelps Road, and Devereux Creek. Data points are represented by open circles, with some points highlighted in blue.

| Site                   | log + 1 transformed average abundance/L          |
|------------------------|--------------------------------------------------|
| North of Venoco Bridge | 0.0, 0.0, 0.5, 1.9, 1.9, 2.9, 5.9, 6.8           |
| East Channel           | 0.4, 0.4, 2.2, 3.4, 3.5, 3.6, 3.6, 4.0, 4.5, 4.6 |
| Main Channel           | 0.0, 0.0, 0.2, 0.3, 4.6                          |
| South of Phelps Bridge | 0.0, 0.5, 0.5, 2.1, 2.2, 2.3, 3.2, 3.3, 4.0, 6.1 |
| North of Phelps Bridge | 0.1, 0.2, 0.3, 3.1, 3.3                          |
| Phelps Road            | 0.4, 1.5, 1.5, 2.0, 4.0                          |
| Devereux Creek         | 0.1, 2.0, 2.6, 3.3, 3.8, 3.8                     |